Finesse × Citadel — Round 2 Report

A Price-Only Momentum / Low-Volatility Portfolio, and an Honest Account of Where Its Return Comes From

Universe: Nifty 100 + Nifty Midcap 100 + Nifty Smallcap 100 (300 names) · **Capital:** ₹1,00,00,000 · **Backtest:** 1 Jan 2021 – 31 Dec 2025 · **Held-out stress test:** 1 Jan – 30 Jun 2026 · **Cost:** 0.1% per transaction, both legs

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1. Problem and Strategy Overview

The mandate is to build and manage a ≤10-stock equity portfolio from the Nifty 100 / Midcap 100 / Smallcap 100 universe, starting from ₹1 crore, and to be ranked on **Total Net PnL** — with a forward-looking out-of-sample test and a jury round that explicitly reward robustness and transparency over hindsight.

Strategy in one sentence: each month, hold the ten most liquid stocks that combine strong 12-month price momentum with low recent volatility, weighting winners more heavily but capping any single name at 25%.

The premium we are harvesting is well documented: **cross-sectional momentum** (past relative winners keep winning over 3–12 month horizons) tempered by a **low-volatility** overlay that trims the drawdowns momentum is prone to. We chose a price-only composite deliberately — momentum and volatility are computable cleanly from daily OHLCV for 300 mid- and small-caps, whereas point-in-time fundamentals for that universe are neither free nor reliable on this timeline. Every design choice is one we can defend from the literature, not one reverse-engineered from the backtest.

Because Total Net PnL over a five-year Indian mid/small-cap bull run will look spectacular for almost *any* long portfolio, the central discipline of this report is separating the return **we created** from the return **the universe handed us**. We quantify that split explicitly (§5) rather than leaving the reader to wonder.

2. Data

Item	Detail
Source	Yahoo Finance via `yfinance`; NSE tickers with `.NS` suffix
Frequency	Daily OHLCV
Period pulled	1 Jun 2019 – 30 Jun 2026 (18-month lead-in so 12-month momentum is computable on the first backtest day)
Features	Split/dividend-adjusted close (returns), raw volume (liquidity screen)
Universe source	Official niftyindices.com constituent lists, snapshot 25 Aug 2026, committed to the repo
Cleaning	`auto_adjust=True` for splits/dividends; forward-

	fill gaps ≤5 days; never fabricate a price before a stock's first trade
Coverage	300 names + 2 benchmark indices; 0 download failures; 0 names above 5% missing days; 64 names listed after 1 Jan 2021 and are simply unscoreable until they have 12 months of history

Two points an evaluator will look for:

- **Adjusted vs raw close.** We use split- and dividend-adjusted close for all return math; using raw close would inject a spurious −50% "return" on every split date. The liquidity screen uses rupee turnover (close × volume) so it is unaffected by the adjustment basis.
- **Survivorship / look-ahead in the universe.** We apply *today's* constituent lists across 2021–2025. This is the compliant reading of the mandate (the rules name the indices with no as-of date and ask that only current-list names be held), but it does bias results upward, because index promotion follows good performance. We do not hide this — we **measure** it (§5) and **stress-test** it against a 2023 point-in-time universe (§7).

3. Methodology

3.1 Stock-selection rule

On the first trading day of each month, using only data up to and including that day:

1. **Liquidity screen** — keep names whose trailing-3-month average daily rupee turnover exceeds **₹5 crore**. A ₹10–25 lakh position is then well under 1% of a typical day's volume, so our fill assumptions are realistic.
2. **Score** each surviving name on a cross-sectional z-score composite: **0.55 × momentum + 0.45 × low-volatility**, where momentum is the cumulative return from t−12 months to t−2 months, and low-volatility is the negative of annualised daily-return σ over the trailing 6 months. Z-scoring each factor first makes the weights meaningful; a name must have a valid value for *both* factors to be scored.
3. **Select** the top 10 by composite score.

3.2 Weighting rule

Rank-proportional among the ten (rank 1..10, so all weights are positive and scale-free — a z-score can be negative, which naive score-proportional weighting cannot handle in a long-only book), then apply a **25% per-name cap** and renormalise iteratively to a fixed point. If the cap cannot be satisfied fully invested, the remainder is held as cash rather than breaching the limit.

3.3 Rebalancing / trading logic

Monthly. **Signal is formed on day *d*'s close and executed at day *d+1*'s close** — one full session of separation, so we never trade on a price we could not have observed when deciding. A **hysteresis buffer** reduces churn: a held name is sold only once it drops out of the top 15, while new entrants must still crack the top 10. The **0.1% cost is charged on the notional of every buy and every sell**, folded into cost basis and proceeds so it flows through realised P&L. Accounting is **share-based** (integer shares + explicit cash; NAV is derived), which is the only way to represent transaction costs and cash drag honestly.

3.4 Risk management

Position cap of 25%; ≤10 names; long-only, cash-funded, never leveraged or short; cash never negative. Beyond the weight cap and the low-volatility leg (which meaningfully tightens drawdown) there are no discretionary overrides — a deliberate choice, so the same rule applies to every holding in every month.

> ***Consistency (mandate §3):*** *the identical scoring, weighting and trading rule is*
>
> *applied to every name in every rebalance. There are no stock-specific adjustments.*

3.5 How the parameters were chosen — and why 2026 played no part

Every parameter is either convention/mandate (12-month lookback, 6-month vol window, 25% cap, 10 names, monthly, ₹5cr screen) or was chosen on an **in-sample train/validate split** — **2021–2023 to choose, 2024–2025 to confirm** — that never touches the 2026 window. Only two parameters were data-chosen at all:

- **Momentum skip = 2 months.** Train selection alpha humps on 2 (+28.9pp), and 2 also validates strongly on 2024–2025 (+11.8pp).
- **Factor weight = 55/45.** Train selection alpha keeps rising to 65–80%, but *validate* peaks at 50–55% and falls above it — the higher train optimum is precisely the region that fails to generalise. We took the validate-optimal 55%, not the train-optimal.

This is the crux of our robustness argument: we selected by **generalisation**, not by in-sample peak, and the out-of-sample window was computed only after the model was frozen. Reproduce with `python -m src.experiments --select`.

4. Tools / Software Used

Python 3. `pandas`/`numpy` for the data panel and vectorised factor math; `scipy` for statistics; `matplotlib` for figures; `pyyaml` for the single config file; `yfinance` for data. No optimiser and no ML — the strategy is a transparent rules engine by design. The backtest engine, metrics and factors are covered by **124 unit tests** (`pytest`), including a no-look-ahead truncation-invariance test. Exact versions in `requirements.txt`; full pipeline in `README.md`.

5. Results and Performance Metrics

5.1 Final portfolio composition (holdings on 31 Dec 2025)

Ticker	Name	Sector	Index	Weight
FORCEMOT	Force Motors	Automobile	Smallcap 100	19.7%
MANAPPURAM	Manappuram Finance	Financial Services	Smallcap 100	17.1%
LTF	L&T Finance	Financial Services	Midcap 100	14.1%
EICHERMOT	Eicher Motors	Automobile	Nifty 100	12.3%
RBLBANK	RBL Bank	Financial Services	Smallcap 100	10.7%
MARUTI	Maruti Suzuki	Automobile	Nifty 100	8.8%

FORTIS	Fortis Healthcare	Healthcare	Midcap 100	6.6%
TVSMOTOR	TVS Motor	Automobile	Nifty 100	5.2%
LAURUSLABS	Laurus Labs	Healthcare	Midcap 100	3.7%
MFSL	Max Financial Services	Financial Services	Midcap 100	1.7%

5.2 Required metrics

Metric	Definition used	Backtest 2021–2025	OOS 2026 H1
Total Net PNL	Final NAV − ₹1 crore	**₹8.99 Cr**	**₹11.93 L**
Final portfolio value		₹9.99 Cr	₹1.12 Cr
Absolute / total return		899.3%	11.9%
Annualised return	Geometric (CAGR)	60.0%	26.5%
Maximum drawdown	Largest peak-to-trough	−29.4%	−17.1%
Sharpe ratio	CAGR ÷ σ(daily)·√252, rf = 0%	2.42	0.93
Sortino ratio	vs downside deviation	3.30	1.39
Gain-to-loss ratio	Avg win ÷ avg loss, per closed position	1.95	1.60
Accuracy	% of closed positions profitable	61.6%	43.8%
— per closing transaction	(finer view, incl. trims)	67.5%	45.2%
Total trades (closed positions)		159	16
Executions (buys / sells)		758 (364/394)	75 (44/31)
Trades per stock	closed positions ÷ names traded	1.38	0.67
Annualised turnover	traded notional ÷ 2·avg NAV	4.43×	5.95×
Total transaction costs paid		₹20.25 L	₹57,907

A "trade" is one closed position (buy → optional trims → full exit), P&L against average cost and net of both legs; open positions are excluded from accuracy. We report the per-transaction view alongside for full transparency.

5.3 The number that actually matters — return decomposition

A five-year Indian mid/small-cap bull market flatters everyone, so we decompose the excess over the Nifty 100 into what the *universe* gave us and what the *strategy* added, using a skill-free equal-weight hold of our own 300-name universe as the fair, bias-matched yardstick (`python -m src.diagnostics`):

	In-sample	Out-of-sample
Excess over Nifty 100	+46.0 pp/yr	+39.4 pp/yr
…universe/composition (size + equal-weight + survivorship — **not skill**)	+20.5 pp/yr	+16.7 pp/yr
…**factor selection (our real contribution)**	**+25.5 pp/yr**	**+22.6 pp/yr**

The headline 899% is real but roughly half of the *excess* over the index is the universe, not us. What we are proud of is the bottom line: **selection alpha is +25.5pp in-sample and +22.6pp out-of-sample** — nearly identical across a genuinely held-out window, which is the strongest evidence the edge is a process and not a fit.

5.4 Figures

`reports/figures/`: equity curve vs benchmark, drawdown, weights-over-time and a monthly-return heatmap, for both windows — all readable in greyscale.

6. Benchmark Comparison

We report **Nifty 100 as the headline benchmark and Nifty 500 as a secondary**, and argue the choice rather than flattering ourselves. No single published index matches a large+mid+small universe. The Nifty 100 is the *harder, more conservative* comparison: it is large-cap, so in a period when small-caps rallied it sets a high bar our mid/small tilt must clear on genuine selection, not on cap drift. Nifty 500 is the honest broad-market comparison (it spans all three cap segments); a literal 1/3-1/3-1/3 blend of the three source indices is unavailable because Yahoo publishes no working Smallcap 100 symbol.

	Portfolio (IS)	Nifty 100 (IS)	Nifty 500 (IS)	Portfolio (OOS)	Nifty 100 (OOS)
Total return	899.3%	89.4%	107.2%	11.9%	−6.4%
Annualised return	60.0%	13.9%	16.0%	26.5%	−12.9%
Max drawdown	−29.4%	−17.6%	−18.8%	−17.1%	−15.0%
Sharpe	2.42	0.98	—	0.93	−0.75
Information ratio	2.40			2.18	
Beta / annualised alpha	1.11 / +44.5%			1.32 / +43.5%	

The strategy outperforms on both return and risk-adjusted return in both windows, with an information ratio above 2. The honest caveat carries over from §5.3: some of the outperformance vs the *large-cap* Nifty 100 is our mid/small-cap tilt, which is why we lead with the bias-matched decomposition rather than this table. Out-of-sample, the portfolio returned +11.9% while both benchmarks fell — the cleanest evidence of downside resilience (down-capture 1.01, i.e. it did not amplify the market's fall).

7. Limitations / Discussion

- **The universe effect is large and disclosed.** ~+20 pp/yr of the in-sample excess is survivorship + size + equal-weight premium, not skill (§5.3). It applies to every team and is held constant on both sides of our selection-alpha measure.
- **Survivorship, isolated.** Re-running on an **August-2023 point-in-time universe** (78% of the Smallcap 100 has since changed) with parameters frozen on ≤2025 data and traded through H1 2026 gives **₹6.94 L profit and +13.5 pp/yr selection alpha in a window where the Nifty 100 fell**

12.9% — with *no* survivorship advantage available. Selection alpha is close to invariant to the universe (+12–13pp clean vs +17–23pp on the current list), which is the whole thesis. (`python -m src.pit_universe`.)

- **Short OOS window.** 122 days and 16 closed positions cannot separate skill from luck on their own — an alphabetical zero-skill control scored +48% over the same window. Trust the *stability* of selection alpha across windows, not any single OOS level.
- **Tuning risk is bounded.** Only two parameters were data-chosen, both on a train/validate split that never saw 2026, both selected for generalisation; the conventional fallbacks (skip=1, 50/50) also validate positive.
- **Momentum's regime dependence.** Momentum crashes hardest when a long downtrend snaps back sharply; a 200-day regime filter was tested to hedge this and *rejected* because it whipsawed (it hurt OOS). The 45% low-volatility leg is our structural, always-on hedge instead.
- **Cost and liquidity modelling.** 0.1% models explicit cost only, not spread, impact or STT; whole-share trading leaves small cash residuals; smallcap fills may cost more than modelled even after the ₹5cr screen.
- **Price-only.** No fundamental or value leg, by design given the data constraints.

Reproducibility

`pip install -r requirements.txt` → `python -m src.data_loader` → `python run_backtest.py`. Selection evidence: `python -m src.experiments --select`. Decomposition: `python -m src.diagnostics`. Robustness search: `python -m src.alt_strategies`. Point-in-time test: `python -m src.pit_universe`. Engine correctness: `pytest` (124 tests). Every number in this report regenerates from the committed code and the single `config.yaml`.